



DTE Platform — V1 Domain Specification (Canonical)



Purpose. This is the canonical, implementation-ready domain specification for **V1 only**. It is the single source of business truth from which schema, contracts, services, screens, and workflows are later derived. It does not redesign the product, add features, or expand scope.

Sources of truth (final). Architecture Blueprint, Domain Decomposition (DDD), Product & Domain Blueprint (DTE-Type-First), and the Ruthless Scope Cut (V1 → V3). The Scope Cut is treated as final.

Boundary. Business/domain layer only. No technology, databases, APIs, services, frameworks, infrastructure, or deployment. Where a downstream choice is needed, it is surfaced as an Open Question, never decided here.

V1 in one sentence. A single issuer can compose, validate, sign, transmit, and prove **FE** (01) and **CCF** (03) documents — surviving an internet outage via contingency, correcting mistakes via plain invalidation — producing immutable evidence and a tamper-evident audit trail.

0. Scope Fence (what V1 is and is not)

In V1 (must exist to issue a single legal document):

- Two document types only: **FE** (01) and **CCF** (03).
- Single issuer, single deployment, manual (human-driven) issuance through a UI.

- Composition → validation → signing → transmission → acceptance tracking.
- Minimal catalog: counterparties and items.
- Plain invalidation (anulación without replacement).
- Minimal contingency (issue offline, transmit later within the legal window).
- Immutable evidence and a hash-chained audit trail.
- Minimal access: two roles, **Owner** and **Biller**.

Explicitly NOT in V1 (deferred per the Scope Cut; named here only to fence them out): API/programmatic issuance, webhooks, batch operations, approval workflows, the substitution saga, replacement-type invalidation, any DTE type beyond FE/CCF, receivables/payments/settlement, inbound reception of supplier documents, email/portal delivery, the generic rule engine, multi-issuer, and full role-based access control. No behavior for these may be invented downstream.

1. Ubiquitous Language (Final)

This vocabulary is canonical. Every other section, and every downstream artifact, must use these exact terms with these exact meanings. No synonyms.

Term	Canonical meaning (no ambiguity)
DTE	Documento Tributario Electrónico. The umbrella legal concept for an electronic fiscal document recognized by the tax authority.
DTE Type	A first-class classification of a fiscal document that determines its rules, identity format, tax behavior, and required parties. In V1 exactly two exist: FE (01) and CCF (03). Never modeled as a mere attribute of an invoice.
FE (Factura Electrónica)	DTE Type 01 . Consumer-facing sale. Receptor identification is optional/relaxed; no fiscal credit is conferred.
CCF (Comprobante de Crédito Fiscal)	DTE Type 03 . Issued to another registered taxpayer; confers IVA fiscal credit. Requires a fully identified taxpayer receptor.
Fiscal Document	The aggregate instance of a single DTE of a specific DTE Type, with its own identity, parties, lines, totals, status, and evidence. The central object of the domain.

Term	Canonical meaning (no ambiguity)
Issuer (Emisor)	The single registered taxpayer operating the deployment and legally responsible for documents issued. The customer who owns the infrastructure, certificate, and tax-authority credentials.
Receptor	The counterparty receiving the document. For CCF must be a fully identified taxpayer; for FE may be a final consumer with relaxed identity.
Counterparty	Master-data record describing a potential receptor (tax identity, name, address). Reusable across documents. Distinct from the immutable Receptor Snapshot.
Item (Product)	Master-data record for a sellable good or service, with a default unit price and tax behavior. Distinct from the Document Line that copies its data at issuance.
Document Line	A single billed line inside a Fiscal Document: description, quantity, unit price, tax treatment, and computed amounts. Snapshotted from an Item or entered manually.
Generation Code (Código de Generación)	A globally unique identifier (UUID form) generated by the platform that uniquely identifies a Fiscal Document for its entire life, independent of the Control Number.
Control Number (Número de Control)	The human-/authority-readable sequential identifier for a document, formatted per DTE Type and issuing point, drawn gaplessly from a Control Number Series.
Control Number Series	The authority over the monotonic, gapless sequence of correlatives for a given DTE Type and issuing point. Guarantees uniqueness and ordering.
Issuer Snapshot	An immutable copy of the issuer's legal identity captured into a document at signing time. Never recomputed from current issuer data.
Receptor Snapshot	An immutable copy of the receptor's identity captured into a document at signing time. Independent of later edits to the Counterparty.
Schema Version	The version of the authority's document definition that a Fiscal Document was built and signed against. Stamped permanently onto the document.
Signed Payload	The exact, canonical serialized content of a document that was cryptographically signed. The legal body of the document.

Term	Canonical meaning (no ambiguity)
	Immutable.
Transmission Record	The append-only history of attempts to deliver a document (or an invalidation) to the tax authority, including each request, response, and outcome.
Acceptance Stamp (Sello de Recepción)	The authority-issued confirmation token returned when a document is accepted (PROCESADO). Proof of legal reception. Immutable once received.
Contingency Episode	A bounded period during which the authority is unreachable and documents are lawfully issued offline, to be transmitted once connectivity is restored within the legal window.
Invalidation (Anulación)	The act, recorded as an Invalidation Event, of legally voiding a previously accepted document. V1 supports only plain invalidation (rescind / other), with no replacement document.
Invalidation Event	The record of a single invalidation: target document, reason type, justification, and its own transmission and acceptance by the authority.
Evidence Bundle	The append-only collection of immutable artifacts proving what was issued: signed payload, authority responses, acceptance stamp, human-readable representation, and content hashes.
Human-Readable Representation	The presentable rendering of a document (the printable/visual form) preserved as evidence. Generated deterministically from the signed payload.
Audit Event / Audit Entry	An immutable, hash-chained record of a single significant business action (who, what, when, on what, with what outcome). The Audit Trail is the ordered chain of these entries.
Owner	The privileged human role: configures the issuer, manages users and catalog, and can perform every issuance/invalidation action.
Biller	The operational human role: composes, signs, transmits, and invalidates documents, but cannot manage users or change issuer configuration.
Draft	A document in composition: fully editable, not yet identified, signed, or legally meaningful.
Validated	A document that has passed all business and structural rules for its DTE Type and has been assigned its identity, ready to sign. Still

Term	Canonical meaning (no ambiguity)
	pre-legal.
Signed	A document whose canonical payload has been cryptographically signed. Legally formed and permanently immutable.
Submitted	A signed document that has been transmitted to the authority and is awaiting an outcome.
Accepted	A submitted document the authority confirmed (PROCESADO), evidenced by an Acceptance Stamp. Legally valid.
Rejected	A submitted document the authority refused (RECHAZADO) with observations. Terminal; never becomes valid; correction is a new document.
Contingency-Held	A signed document deliberately held for later transmission because it was issued during a Contingency Episode.
Invalidated	A previously accepted document whose Invalidation Event was itself accepted by the authority. Legally void but fully retained.

2. Bounded Contexts (V1 Only)

Only contexts that survive the Ruthless Scope Cut appear. Each states purpose, responsibilities, what it owns, and what it must never own.

2.1 Document Type Registry

- **Purpose.** Hold the canonical specification of each supported DTE Type so that all rules are data-driven through a stable seam, even though only FE and CCF exist in V1.
- **Responsibilities.** Define, per type: required fields, allowed receptor identity rules, tax behavior, control-number format, schema version, and which operations are permitted.
- **Owns.** The Document Type Specification for **01** and **03**; the active Schema Version per type.
- **Must never own.** Document instances, sequence state, transmission, or any logic specific to a single document. It is configuration, not workflow. In V1 it is

read-only at runtime (specs are fixed at deployment).

2.2 Issuance

- **Purpose.** Compose, validate, identify, sign, transmit, and track acceptance of Fiscal Documents. The heart of V1.
- **Responsibilities.** Manage the Fiscal Document lifecycle through its state machine; assign identity from the Control Number Series; capture immutable snapshots; orchestrate signing; create Transmission Records; interpret authority outcomes.
- **Owns.** FiscalDocument, ControlNumberSeries, TransmissionRecord; the document state machine; the issuer/receptor snapshot capture.
- **Must never own.** Counterparty/Item master data (only snapshots of them), the cryptographic key custody, evidence retention rules (it triggers evidence creation but Evidence & Archive owns persistence), or invalidation logic (delegated to Invalidation).

2.3 Invalidation

- **Purpose.** Lawfully void an already-accepted document via plain invalidation.
- **Responsibilities.** Create and transmit an Invalidation Event referencing exactly one accepted document; track its acceptance; drive the target document to Invalidated.
- **Owns.** InvalidationEvent and its own transmission/acceptance lifecycle.
- **Must never own.** Replacement/substitution logic (deferred), modification of the original document's signed payload, or creation of any new Fiscal Document.

2.4 Contingency

- **Purpose.** Allow lawful offline issuance when the authority is unreachable, and reconcile once connectivity returns.
- **Responsibilities.** Open and close a Contingency Episode; mark documents as Contingency-Held; record the contingency justification; ensure the

contingency notification and held documents are transmitted within the legal window.

- **Owns.** ContingencyEpisode and the association of held documents to an episode.
- **Must never own.** Document content, signing, or the decision of what a document contains. It governs timing and mode of transmission, not the document itself.

2.5 Parties & Counterparties

- **Purpose.** Maintain reusable receptor master data.
- **Responsibilities.** Create, edit, and look up Counterparties; validate tax-identity well-formedness for use in CCF vs FE.
- **Owns.** Counterparty records.
- **Must never own.** Receptor Snapshots inside documents (those belong to the document forever), or any document state.

2.6 Product & Catalog

- **Purpose.** Maintain reusable item master data and default pricing.
- **Responsibilities.** Create, edit, and look up Items; hold default unit price and tax behavior.
- **Owns.** Item records.
- **Must never own.** Document Lines (snapshots), price history, or multi-tier price lists (deferred).

2.7 Evidence & Archive

- **Purpose.** Preserve immutable, append-only proof of everything issued.
- **Responsibilities.** Create and store Evidence Bundles and their artifacts; compute and retain content hashes; guarantee artifacts never change.
- **Owns.** EvidenceBundle and its EvidenceArtifacts.

- **Must never own.** Mutable business state, document composition, or the right to alter or delete any artifact.

2.8 Audit Trail

- **Purpose.** Record a tamper-evident history of who did what.
- **Responsibilities.** Append hash-chained Audit Entries for every significant action; expose the chain for verification.
- **Owns.** The Audit Trail (the ordered, hash-linked sequence of Audit Entries).
- **Must never own.** Business decisions, or any ability to edit/delete entries. It observes; it never mutates other contexts.

2.9 Access & Identity

- **Purpose.** Know who the actor is and whether they may perform an action.
- **Responsibilities.** Authenticate human actors; assign exactly one of two roles (Owner, Biller); authorize actions per role.
- **Owns.** User identities and role assignment.
- **Must never own.** Service accounts/API identities (deferred), granular permission matrices, or any document/business data.

3. Aggregate Roots

For each: purpose, invariants, lifecycle, business rules. Aggregate boundaries define consistency: everything inside a root is changed together; cross-aggregate references are by identity only.

3.1 FiscalDocument (root)

- **Purpose.** Represent one DTE of a specific DTE Type through its entire life, from draft to a terminal legal state.
- **Contains (internal).** Document header, DTE Type reference, Generation Code, Control Number, Schema Version, Issuer Snapshot, Receptor Snapshot, Document Lines, Tax Summary, Totals, Operation Condition, Status, and

references (by identity) to its Transmission Records, Evidence Bundle, and any Invalidation Event.

- **Invariants.**

- A document always has exactly one DTE Type, fixed at creation; it can never change type.
- Once Signed, the Signed Payload and all snapshots are immutable.
- A document in Validated or later always has a Generation Code and a Control Number; a Draft has neither.
- Totals always equal the sum of line amounts plus the Tax Summary; the document cannot leave Draft if they do not reconcile.
- A CCF must have a fully identified taxpayer Receptor Snapshot; an FE may not require it.
- Status transitions must follow the state machine (Section 6) — no skips, no reversals beyond those defined.

- **Lifecycle.** Draft → Validated → Signed → (Submitted | Contingency-Held) → Accepted | Rejected; Accepted → Invalidated. Terminal states: Rejected, Invalidated, and Accepted (until/unless invalidated).

- **Business rules.**

- Identity (Generation Code + Control Number) is assigned exactly once, at the transition into Validated, and never reused or regenerated.
- Signing freezes the canonical payload; any later correction is a new FiscalDocument, never an edit.
- Only an Accepted document may be invalidated.
- A Rejected document is terminal and its Control Number is considered consumed (recorded, never reused).

3.2 ControlNumberSeries (root)

- **Purpose.** Be the single authority for gapless, ordered correlatives for one DTE Type at one issuing point.
- **Invariants.**

- Correlatives are strictly monotonic and gapless within the series.
- Every assigned Control Number is globally unique and assigned to exactly one FiscalDocument.
- A correlative, once handed out, is never handed out again — even if the document is later Rejected or Invalidated.
- **Lifecycle.** Created once per (DTE Type, issuing point) at configuration; lives for the deployment's life; only advances.
- **Business rules.**
 - Assignment is atomic: a document either receives the next correlative or none.
 - The series never rewinds, even to fill a gap left by a rejected document; gaps are explained by document records, not by reusing numbers.

3.3 TransmissionRecord (root)

- **Purpose.** Be the append-only truth of every interaction with the authority for a single document or a single invalidation.
- **Contains (internal).** Ordered Transmission Attempts, each with timestamp, purpose (document submission vs invalidation submission vs contingency notification), the exact payload sent, the exact response received, and the interpreted outcome.
- **Invariants.**
 - Records and attempts are append-only; never edited or deleted.
 - Each record references exactly one target (a FiscalDocument or an InvalidationEvent).
 - An Acceptance Stamp, once recorded, is immutable.
- **Lifecycle.** Opened on first transmission attempt; gains attempts; effectively closed when a terminal outcome (accepted/rejected) is recorded, but remains forever readable.
- **Business rules.**
 - Every attempt, including failures and timeouts, must be appended.

- The interpreted outcome must be derived from the authority's response, never inferred optimistically.

3.4 InvalidationEvent (root)

- **Purpose.** Record and execute the lawful voiding of one accepted document.
- **Invariants.**
 - References exactly one FiscalDocument, which must be Accepted at the time of creation.
 - Carries a reason type limited in V1 to rescind or other (no replacement type).
 - Once its own transmission is Accepted, it is immutable and the target becomes Invalidated.
- **Lifecycle.** Created → Submitted → Accepted | Rejected. If Accepted, drives the target document to Invalidated; if Rejected, the target remains Accepted and a new attempt may be made.
- **Business rules.**
 - A document may have at most one Accepted Invalidation Event.
 - An invalidation never alters the original signed payload or evidence; it adds a new fact.

3.5 ContingencyEpisode (root)

- **Purpose.** Govern a bounded period of lawful offline issuance and its later reconciliation.
- **Contains (internal).** Open/close timestamps, contingency reason, the set of Contingency-Held documents (by identity), and the state of the required contingency notification to the authority.
- **Invariants.**
 - An episode has a defined start; it cannot be closed until the contingency notification and all held documents have reached a terminal transmission outcome, or the legal window rule is recorded as breached.

- A document may belong to at most one Contingency Episode.
- **Lifecycle.** Open → Reconciling (connectivity restored, transmitting) → Closed. (Detailed in Section 6.)
- **Business rules.**
 - Documents are issued and signed normally during an episode but are Contingency-Held instead of Submitted.
 - On reconciliation, the contingency notification is transmitted first, then the held documents, within the legal window.
 - The legal window (and exact reconciliation order) is a hard rule — see Open Questions for the precise duration to confirm.

3.6 Counterparty (root)

- **Purpose.** Reusable receptor master data.
- **Invariants.** Tax identity must be well-formed for its kind; a counterparty intended for CCF must carry a complete taxpayer identity.
- **Lifecycle.** Created → edited → (optionally) deactivated. Never hard-deleted if referenced by any document snapshot history.
- **Business rules.** Editing a Counterparty never changes any past Receptor Snapshot.

3.7 Item (root)

- **Purpose.** Reusable product/service master data with default price and tax behavior.
- **Invariants.** Has a non-negative default unit price and a defined tax behavior.
- **Lifecycle.** Created → edited → (optionally) deactivated. Never hard-deleted if referenced by any document line history.
- **Business rules.** Editing an Item never changes any past Document Line.

3.8 EvidenceBundle (root)

- **Purpose.** Hold the immutable proof set for exactly one FiscalDocument.

- **Contains (internal).** Evidence Artifacts: the Signed Payload, each authority response, the Acceptance Stamp, the Human-Readable Representation, and content hashes; plus, when applicable, invalidation evidence.
- **Invariants.**
 - Append-only: artifacts may be added but never modified or removed.
 - Each artifact carries a content hash captured at the moment of addition.
 - The bundle references exactly one FiscalDocument.
- **Lifecycle.** Created when the document is first Signed; grows as evidence accrues (acceptance, invalidation); never closed for reading.
- **Business rules.** Evidence is created as a side effect of irreversible domain facts (signing, acceptance, invalidation) and must be retained even for Rejected/Invalidated documents.

3.9 AuditTrail (root)

- **Purpose.** Tamper-evident chronological record of significant actions across all contexts.
- **Contains (internal).** Ordered Audit Entries, each hash-linked to the previous.
- **Invariants.**
 - Append-only and ordered; entries are never edited or deleted.
 - Each entry's hash incorporates the previous entry's hash, forming a verifiable chain; breaking the chain is detectable.
- **Lifecycle.** Begins at deployment initialization; only grows.
- **Business rules.** Every state transition, signing, transmission outcome, invalidation, contingency open/close, and access-control decision worth defending in an inspection must produce an entry.

3.10 User (Access root)

- **Purpose.** Represent a human actor and their single role.
- **Invariants.** Exactly one role per user (Owner or Biller); at least one Owner must always exist.

- **Lifecycle.** Created → role assigned → (optionally) deactivated.
- **Business rules.** Only an Owner manages users and issuer configuration; both roles may issue and invalidate documents.

3.11 DocumentTypeSpecification (Registry root)

- **Purpose.** Hold the fixed rule set for a DTE Type behind a stable seam.
- **Invariants.** Exactly the two specifications `01` and `03` exist in V1; each pins a Schema Version, required fields, receptor identity rule, tax behavior, and control-number format.
- **Lifecycle.** Defined at deployment; read-only at runtime in V1.
- **Business rules.** All type-specific validation in Issuance reads from here; no type rule is hard-coded outside this seam, so adding types later requires no redesign.

4. Entities

Entities have identity and a lifecycle but live inside an aggregate root (they are not independently consistent).

- **Document Line.**
 - *Meaning.* One billed good/service within a Fiscal Document.
 - *Identity.* A line identifier unique within its document.
 - *Lifecycle.* Created/edited freely while the document is Draft; frozen at signing.
 - *Relationships.* Belongs to one FiscalDocument; snapshots data from one Item (or manual entry); contributes to the Tax Summary and Totals.
- **Transmission Attempt.**
 - *Meaning.* A single send/receive interaction with the authority.
 - *Identity.* Sequence position within its Transmission Record.

- *Lifecycle*. Created on send; finalized when a response (or failure) is recorded; never altered after.
- *Relationships*. Belongs to one TransmissionRecord, which targets one FiscalDocument or InvalidationEvent.
- **Evidence Artifact.**
 - *Meaning*. One immutable proof item (signed payload, response, stamp, representation, hash).
 - *Identity*. Artifact identifier unique within its bundle, plus a kind.
 - *Lifecycle*. Created once and never changed.
 - *Relationships*. Belongs to one EvidenceBundle, which belongs to one FiscalDocument.
- **Audit Entry.**
 - *Meaning*. One recorded significant action.
 - *Identity*. Monotonic position in the chain plus its own hash.
 - *Lifecycle*. Appended once; immutable.
 - *Relationships*. Links to the previous entry by hash; references the subject (document, user, episode, etc.) by identity.
- **Contingency Membership.**
 - *Meaning*. The fact that a specific document was issued under a specific Contingency Episode.
 - *Identity*. The (episode, document) pair.
 - *Lifecycle*. Created when the document is held; resolved when the document reaches a terminal transmission outcome.
 - *Relationships*. Connects one ContingencyEpisode to one FiscalDocument.
- **Issuing Point.**
 - *Meaning*. The establishment/point-of-sale coordinate that, with DTE Type, scopes a Control Number Series. In V1 a single issuing point may suffice

but the concept must exist so numbering and control-number format are correct.

- *Identity*. Establishment + point-of-sale codes.
- *Lifecycle*. Configured by Owner; stable.
- *Relationships*. Scopes one or more ControlNumberSeries; stamped into each Control Number.

5. Value Objects

Value objects are immutable, identity-less, and compared by value. They cannot be edited in place — only replaced (and only on a mutable aggregate, never after signing).

- **DTE Type**. Code (`01` or `03`), canonical name, and bound Schema Version. Determines rules; immutable on a document.
- **Generation Code**. Globally unique UUID-form identifier; assigned once; never changes.
- **Control Number**. Formatted sequential identifier composed of a DTE-Type segment, an Issuing Point segment, and a gapless correlative. Conceptual form: `DTE-<type>-<issuingPoint>-<correlative>` (for example `DTE-01-M001P001-0000000000000001`). Unique; immutable once assigned.
- **Schema Version**. The authority document-definition version a document was built and signed against; stamped permanently.
- **Tax Identity**. A discriminated identity: kind (taxpayer registration, national tax number, or consumer document) plus its value, validated for well-formedness. CCF requires the taxpayer-grade identity; FE may accept a relaxed or absent one.
- **Address**. Structured location (department, municipality, detail line). Used inside snapshots.
- **Money**. Amount plus currency (USD in V1), with fixed-scale decimal semantics; non-negative for prices, signed only where the domain defines it.

- **Quantity.** Non-negative measured amount with a unit.
 - **Unit Price.** Money per unit, non-negative.
 - **Tax Line.** A tax category (for example standard IVA at 13%, or exempt/non-subject) and the computed tax amount for the base it applies to.
 - **Tax Summary.** The aggregated set of Tax Lines and the resulting tax totals for a document.
 - **Document Totals.** Net, tax, and gross totals derived from lines and the Tax Summary; must reconcile.
 - **Operation Condition.** Cash or credit terms of the sale (descriptive in V1; no receivable is created).
 - **Document Status.** Enumerated lifecycle state (Draft, Validated, Signed, Submitted, Contingency-Held, Accepted, Rejected, Invalidated).
 - **Issuer Snapshot.** Immutable captured issuer identity (legal name, tax identity, address, issuing point) at signing.
 - **Receptor Snapshot.** Immutable captured receptor identity at signing, independent of the Counterparty thereafter.
 - **Acceptance Stamp.** The authority's reception token plus its timestamp; immutable proof of acceptance.
 - **Authority Response.** The interpreted outcome of a transmission: status (accepted, rejected, or a contingency/availability signal), observations, and timestamp.
 - **Invalidation Reason.** The reason type (rescind or other) plus a free justification; constrained to V1's non-replacement set.
 - **Contingency Reason.** The cause/category of an episode (for example, authority unreachable) plus justification.
 - **Content Hash.** A cryptographic digest of an artifact's exact bytes, captured at creation, enabling tamper detection.
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6. State Machines

Transitions are exhaustive. Any transition not listed is forbidden. Each transition lists trigger, preconditions, and resulting state.

6.1 FiscalDocument

From → To	Trigger	Preconditions	Result / effects
(none) → Draft	Create document of a chosen DTE Type	Actor authorized; DTE Type is 01 or 03	Empty editable document; no identity yet
Draft → Draft	Edit lines/receptor/condition	Status is Draft	Content updated; totals recomputed
Draft → Validated	Validate document	All type rules pass; receptor satisfies type; totals reconcile	Generation Code and Control Number assigned; snapshots prepared
Validated → Draft	Reopen for edit	Status is Validated; not yet Signed	Identity is retained-but-held; see rule below
Validated → Signed	Sign document	Status is Validated; identity present; signing succeeds	Signed Payload frozen; snapshots frozen; Evidence Bundle created
Signed → Submitted	Transmit to authority	Status is Signed; no open Contingency Episode	Transmission Record opened; awaiting outcome
Signed → Contingency-Held	Issue under contingency	Status is Signed; an open Contingency Episode exists	Document joined to the episode; transmission deferred
Contingency-Held → Submitted	Reconcile after connectivity	Episode is Reconciling; contingency notification accepted	Transmission Record opened; awaiting outcome

From → To	Trigger	Preconditions	Result / effects
Submitted → Accepted	Authority returns accepted	Outcome is accepted with a stamp	Acceptance Stamp stored in evidence; document legally valid
Submitted → Rejected	Authority returns rejected	Outcome is rejected with observations	Observations stored; document terminal; correction is a new document
Accepted → Invalidated	Invalidation Event accepted	An InvalidationEvent for this document is Accepted	Document legally void; all evidence retained

Rules binding the machine:

- Identity is assigned at Draft → Validated and never regenerated. If a Validated document is reopened to Draft, its assigned correlative is considered consumed; re-validation draws a new correlative (the previous one is recorded as consumed, preserving gaplessness). This is the safe default; see Open Questions if reuse-on-reopen is desired instead.
- Signing is irreversible. There is no transition out of Signed except Submitted or Contingency-Held.
- Rejected and Invalidated are terminal. Accepted is terminal unless invalidated.
- A document may not be Submitted while any Contingency Episode is open.

6.2 InvalidationEvent

- **(none) → Created:** trigger = start invalidation; precondition = target document is Accepted and not already invalidated; result = event drafted with reason type and justification.
- **Created → Submitted:** trigger = transmit invalidation; precondition = event complete; result = transmission opened.
- **Submitted → Accepted:** trigger = authority accepts; result = target document → Invalidated; invalidation evidence stored.

- **Submitted** → **Rejected**: trigger = authority rejects; result = target remains Accepted; a new invalidation attempt may be created.

6.3 ContingencyEpisode

- **(none)** → **Open**: trigger = contingency declared (authority unreachable); precondition = no other open episode; result = subsequent signed documents are Contingency-Held.
- **Open** → **Reconciling**: trigger = connectivity restored / reconciliation started; precondition = episode open; result = contingency notification transmitted first.
- **Reconciling** → **Closed**: trigger = all held documents reached a terminal transmission outcome and the contingency notification was accepted; precondition = nothing pending; result = episode closed; no new documents may join.

7. Invariants — the constitution of the domain

These must always be true. Treat any violation as a domain-level failure, never an acceptable edge case.

Identity & numbering

1. Every Validated-or-later document has exactly one Generation Code and exactly one Control Number.
2. Control Numbers are globally unique and never reused.
3. Control Number Series are gapless and monotonic; a consumed correlative is never re-issued, including after rejection, reopening, or invalidation.
4. A document's DTE Type is fixed at creation and never changes.

Immutability

1. A Signed document's Signed Payload is immutable forever.
2. Issuer Snapshot and Receptor Snapshot are immutable once signed and are never recomputed from current master data.

3. Snapshots never change when their source Counterparty or Item is later edited.
4. An Acceptance Stamp, once recorded, is immutable.

Evidence & audit

1. Evidence Bundles are append-only; artifacts are never modified or deleted, even for Rejected or Invalidated documents.
2. Every Evidence Artifact carries a content hash captured at creation.
3. The Audit Trail is append-only and hash-chained; any tampering is detectable.
4. Audit Entries cannot be edited or deleted.

Lifecycle integrity

1. State transitions occur only as defined in Section 6; no skips or undefined reversals.
2. Only an Accepted document may be invalidated, and at most one Accepted Invalidation Event may exist per document.
3. Correction of a Rejected or Invalidated document is always a new FiscalDocument; existing documents are never resurrected or edited.
4. A document may belong to at most one Contingency Episode.

Consistency of content

1. Document Totals always reconcile with the sum of Document Lines and the Tax Summary before a document may leave Draft.
2. A CCF always carries a complete taxpayer Receptor Snapshot; an FE follows its relaxed receptor rule.
3. Every transmission attempt — success, rejection, or failure — is recorded; outcomes are derived only from the authority's actual response.

Access

1. Every state-changing action is attributable to exactly one authenticated actor, and at least one Owner always exists.

8. Domain Events

Each event states when it occurs, why it exists, and its business meaning. Events are facts in the past tense and are themselves auditable.

Event	When it occurs	Why it exists / business meaning
DocumentCreated	A new Draft is started	Marks the beginning of a document's life; anchors all later facts
DocumentValidated	A Draft passes all rules and is identified	Identity (Generation Code + Control Number) is now bound; ready to sign
DocumentSigned	The canonical payload is cryptographically signed	The document is legally formed and immutable; evidence begins
DocumentSubmitted	A signed document is transmitted to the authority	The issuer has asserted the document to the authority; awaiting judgment
DocumentAccepted	The authority confirms with a stamp	The document is legally valid; the strongest proof of issuance
DocumentRejected	The authority refuses with observations	The document is void on arrival; signals a correction is needed (as a new document)
ContingencyStarted	A Contingency Episode is opened	Lawful offline issuance begins; documents will be held
DocumentHeldUnderContingency	A signed document joins an open episode	Records that legality depends on later timely transmission
ContingencyNotified	The contingency notification is accepted by the authority	The lawful basis for the held documents is registered; held documents may now transmit

Event	When it occurs	Why it exists / business meaning
ContingencyClosed	All held documents reach a terminal outcome	The episode is fully reconciled; no further holds
InvalidationRequested	An Invalidation Event is created for an accepted document	The issuer intends to void a valid document; reason captured
DocumentInvalidated	The Invalidation Event is accepted by the authority	The document is legally void while fully retained as evidence
InvalidationRejected	The authority refuses an invalidation	The target remains valid; a new attempt may follow
EvidenceArtifactStored	Any immutable artifact is added to a bundle	Guarantees a durable, hashed record of a legal fact
TransmissionAttempted	Any send to the authority, success or failure	Complete, honest record of integration behavior for troubleshooting and proof

9. Audit Requirements

The audit trail must let a future inspector, troubleshooter, or court reconstruct exactly what happened, by whom, and when — without trusting the current state of mutable data.

Every Audit Entry records:

- *Who*. The authenticated actor (user identity and role) responsible for the action.
- *What*. The action taken, in domain terms (validate, sign, submit, accept-recorded, reject-recorded, invalidate, contingency open/close, user/role change, configuration change).
- *When*. A trustworthy timestamp.

- *On what.* The subject identity (Generation Code / Control Number, Invalidation Event, Contingency Episode, User, configuration key).
- *Outcome.* Success or failure and any authority-returned status or observation.
- *Integrity link.* The hash of the previous entry and this entry's own hash.

Actions that must always be audited:

- Every FiscalDocument state transition (Section 6.1).
- Every signing operation (including failures).
- Every transmission attempt and its interpreted outcome.
- Every invalidation request, acceptance, and rejection.
- Every contingency episode open, notify, and close.
- Every access-control decision that grants or denies a state-changing action.
- Every change to issuer configuration, users/roles, and catalog master data (create, edit, deactivate).

Audit guarantees:

- Append-only and hash-chained (Invariants 11–12).
- Never blocks on, nor is blocked by, business success — a failed action is still audited.
- Sufficient on its own to explain a document's full history even if no other context were consulted.

10. Evidence Preservation Rules

What evidence exists (per FiscalDocument, in its Evidence Bundle):

- The exact Signed Payload (the legal body that was signed).
- Every Authority Response received (acceptance, rejection, contingency/availability signals).
- The Acceptance Stamp, when accepted.

- The Human-Readable Representation generated deterministically from the signed payload.
- A Content Hash for each artifact.
- For invalidated documents: the Invalidation Event payload, its authority response, and its acceptance stamp.

When evidence is created:

- On **DocumentSigned**: the bundle is created and the Signed Payload + its hash + the Human-Readable Representation are stored.
- On each transmission outcome: the Authority Response (and, if accepted, the Acceptance Stamp) is stored.
- On invalidation acceptance: the invalidation evidence is appended.

Whether evidence can change:

- Never. Evidence Bundles are append-only; artifacts are immutable once stored (Invariants 9–10). Correction does not edit evidence — it produces a new document with its own bundle.
- Evidence is retained in full even for Rejected and Invalidated documents; nothing is deleted.

Retention assumptions:

- Evidence must be retained for at least the legal preservation period applicable to fiscal documents (a multi-year horizon; the exact duration is confirmed in Open Questions).
- The Human-Readable Representation must remain reproducible/legible for the full retention period independent of any rendering technology.
- Evidence is preserved at issuance time precisely because it cannot be reconstructed later; this is why evidence and the audit chain are V1, not deferred.

11. Open Questions (implementation-blocking only)

Only questions that genuinely block building V1 business behavior. No enhancements, no V1.5+ ideas.

1. **Contingency legal window.** What is the exact maximum time, after connectivity is restored, to (a) transmit the contingency notification and (b) transmit each held document? The state machine enforces ordering, but the concrete durations must be confirmed to encode the deadline rule.
2. **Correlative on reopen.** When a Validated document is reopened to Draft, is the assigned correlative permanently consumed (current safe default) or reusable by that same document on re-validation? This changes the gaplessness rule's edge behavior.
3. **Rejected-document numbering.** Does a Rejected document leave a permanent gap in the series (current assumption), or must its correlative be formally accounted for differently to satisfy the authority's gapless expectations?
4. **FE receptor minimum.** What is the precise minimum receptor identity an FE may carry (fully anonymous final consumer vs. a required threshold for higher amounts)? This determines the FE validation rule in the Document Type Registry.
5. **CCF receptor completeness.** What is the exact set of receptor fields the authority requires for a CCF to be accepted, so validation rejects incomplete CCFs before signing rather than after authority rejection?
6. **Schema Version source.** Which authority schema version(s) are pinned for 01 and 03 at first commercial install, and what is the business rule when the authority publishes a new version mid-operation (V1 assumes the pinned version is fixed at deployment)?
7. **Tax treatments in scope.** Beyond standard IVA at 13%, which tax line categories (exempt, non-subject) must V1 support for FE/CCF lines, so the Tax Summary rules are complete?
8. **Retention duration.** What is the confirmed minimum legal retention period for evidence and audit data, to set the retention rule precisely?
9. **Invalidation time limit.** Is there a deadline after acceptance beyond which a document can no longer be invalidated? If so, it becomes a precondition on

InvalidationEvent creation.



This V1 Domain Specification is complete, technology-agnostic, and aligned to the Ruthless Scope Cut. An implementation team can derive schema, contracts, services, screens, and workflows from it without inventing business behavior — pending only the nine Open Questions, each of which is a factual rule to confirm, not a design decision to make.